

1. Soil Health Diagnosis & Nutrient Status

Overall Health Score	Status Assessment	Identified Critical Deficiencies
63 / 100	Moderate Depletion Requires restorative legume rotation	■ Low Nitrogen (42 kg/ha vs 80 target) ■ Low Organic Carbon (0.52% vs 0.8% target)

Parameter	Measured Value	Optimal Range	Status & Impact
Nitrogen (N)	42.0 kg/ha	80 - 160 kg/ha	Deficient - Continuous tomato harvest depleted soil N
Phosphorus (P)	28.0 kg/ha	30 - 60 kg/ha	Sub-optimal - Root development restricted
Potassium (K)	55.0 kg/ha	60 - 120 kg/ha	Adequate - Good fruit quality potential
Soil pH	6.50	6.0 - 7.5	Optimal - Ideal nutrient solubility
Organic Carbon	0.52%	0.80 - 1.50%	Deficient - Low microbial biomass & water retention

2. Monoculture Warning & Risk Assessment

■ CONTINUOUS CULTIVATION PENALTY DETECTED (Tomato × 3 Consecutive Seasons)

Tomato monoculture has depleted solanaceous micronutrients and caused early-blight pathogen spore buildup in soil. A strict rotation restriction has been applied: **Tomato is penalized by -30%** for the upcoming season to force biological recovery.

3. Recommended 3-Season Restorative Crop Rotation

Season 1 (Immediate)	Season 2 (Restoration)	Season 3 (Stabilization)
Green Gram (Moong) - Family: Legume (N-Fixer) - Duration: 75 Days - Market Price: ■85/kg (Mandi) - Expected Profit: ■33,500/acre - Role: Fixes 35-45 kg N/ha into soil nodules	Groundnut - Family: Legume (Restorer) - Duration: 105 Days - Market Price: ■72/kg (Mandi) - Expected Profit: ■41,200/acre - Role: Deep root aeration & organic matter	Wheat / Maize - Family: Cereal - Duration: 115 Days - Market Price: ■23.75/kg (Mandi) - Expected Profit: ■27,300/acre - Role: High biomass feeder utilizing fixed N

4. Projected Financial Returns & Soil Health Recovery Trajectory

Stage	Crop Cultivated	Projected Net Profit / Acre	Farm Total (4.5 Ac)	Soil Health Score Trajectory
Current Baseline	Depleted Tomato Soil	—	—	58 / 100 (Baseline)
Season 1	Green Gram (Moong)	■33,500	■1,50,750	65 / 100 (+7 recovery)
Season 2	Groundnut	■41,200	■1,85,400	72 / 100 (+7 recovery)
Season 3	Wheat / Maize	■27,300	■1,22,850	79 / 100 (Fully Restored)
3-Season Total	Optimized Rotation	■1,02,000 / acre	■4,59,000	79 / 100 (Optimal)

■ Key Farmer Action Items:

- Inoculate Rhizobium:** Coat Green Gram seeds with Rhizobium bio-fertilizer to maximize atmospheric N-fixation.
- Reduce Chemical Urea by 30%:** Legume root nodules naturally enrich 40 kg N/ha, saving input fertilizer cost.
- Mulch Crop Residues:** Retain crop residues after harvest to restore organic carbon above the 0.8% threshold.